

# HAT-P-55b

R-0943

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-55\_260624 · created 2026-07-29T13:15:40+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2461215.952 +/- 0.011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.145 +/- 0.047               |
| Transit depth (Rp/Rs)^2                           | 2.11 +/- 1.37 [%]             |
| Orbital Inclination (inc)                         | 87.54 +/- 2.38                |
| Airmass coefficient 1 (a1)                        | 0.45 +/- 0.19                 |
| Airmass coefficient 2 (a2)                        | 0.56 +/- 1.71                 |
| Scatter in the residuals of the lightcurve fit is | 38.57 %                       |
| Best Comparison Star                              | None                          |
| Optimal Aperture                                  | 2.53                          |
| Optimal Annulus                                   | 6.0                           |
| Transit Duration (day)                            | 0.083 +/- 0.018               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.145 ± 0.047 vs 0.1202 (deviation 0.46σ) |
| Duration fit vs published | 0.083 d vs 0.1225 d (deviation 1.9σ)      |
| Mid-time O–C              | -12.3 min                                 |
| Depth SNR                 | 2.7                                       |
| Residual scatter          | 38.57 %                                   |

### Light curve

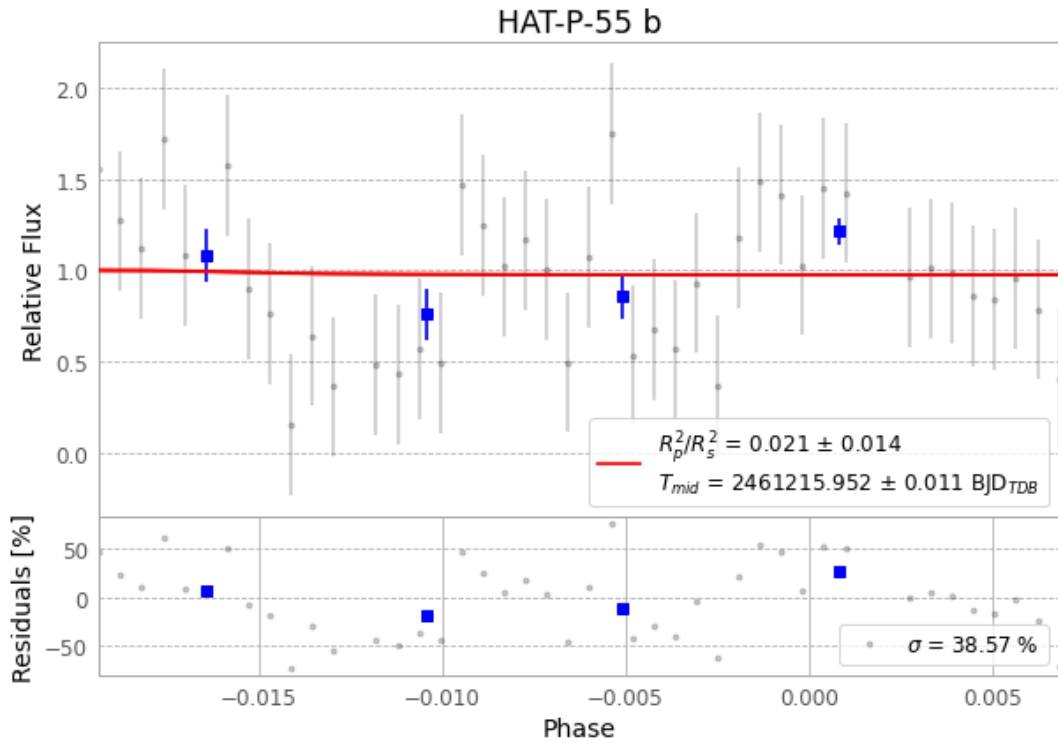


Figure 1 — FinalLightCurve\_HAT-P-55 b\_2026-06-24.png (SHA-256 969588ffc3bccf65...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [504, 332], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b82d5a888c0f294e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

46 FITS frames (30 MB), HATP-55260624090315.FITS → HATP-55260624111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-260624.log (de2fbd1f9536...), FinalLightCurve\_HAT-P-55 b\_2026-06-24.png (969588ffc3bc...), FinalLightCurve\_HAT-P-55 b\_2026-06-24.csv (ffa007a9662c...)

## Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 13:15Z.